

**Determining Solutions to Inequalities**

Worksheet 7A.01.08

Name \_\_\_\_\_

Date \_\_\_\_\_

Period \_\_\_\_\_

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**Use this table of possible values for questions 1 and 2.**

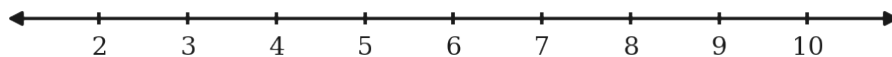
|       |    |                |                  |       |
|-------|----|----------------|------------------|-------|
| -10.5 | 0  | $6\frac{3}{8}$ | -4.999           | -20.6 |
| 42    | -5 | 12.75          | $-1\frac{6}{17}$ | -7.25 |

1. Which values from the table are solutions to  $-25 < 4x - 5$  ?
2. Which values from the table are solutions to  $-25 > 4x - 5$  ?
3. Rewrite both inequalities so that  $-5$  is a solution to each. Explain your reasoning.

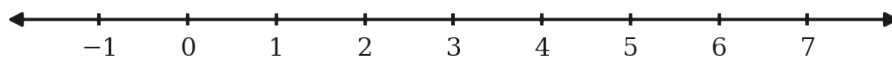
4. Complete the sentence. An inequality has \_\_\_\_\_ solutions.
5. Complete the sentence. When you multiply or divide both sides of an inequality by a negative number, the inequality symbol \_\_\_\_\_.
6. Tanner and Jonah were solving the inequality  $-3 - y/7 < -5$ . Jonah says the solution is  $y < -14$ . Tanner says the solution is  $y > -14$ . Do you agree with one of them, or with neither? Justify your answer mathematically.

For questions 7 through 10, solve the inequality, then graph the solution set on the number line provided.

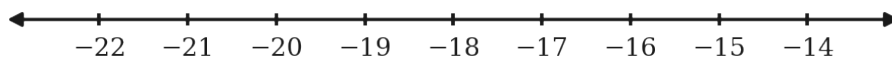
7.  $5 + \frac{2}{3}b \geq 9$



8.  $4.6c - 9.5 < 4.3$



9.  $-1 \leq -4 - y/6$



10.  $-7x - 12 > 16$

